4W1H Keyword Extraction based Summarization Model

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***Abstract*—In this internet era, with rapidly growing online information, there is a need for automatic summarization of textual documents from plethora of available information, making it an interesting area of research. Automatic keyword extraction and text summarization are Natural Language Processing (NLP) tasks for extracting relevant information from the large text documents. 4W1H (Who, When, Where, What, How) keywords are crucial for sentence generation. Despite the potential of 4W1H keywords, there have not been approaches that utilize the keywords in NLP tasks, particularly summarization. In this paper, we propose a new summarization method based on 4W1H keywords extraction which extracts the answer to a question corresponding to each event in QA format. We apply our methods to BERT and ELECTRA models to generate a summary, which are well-known pre-trained Language Models (LMs) in NLP domain, as a baseline. In experiments, our 4W1H keyword extraction method shows promising performance on AI Hub* Machine Reading Comprehension (MRC) dataset, recording an extraction performance of an F1-score as 84.93%. Moreover, we show the results of generating a rule-based summarization using keywords extracted with 4W1H.**

***Keywords*—Natural Language Processing, Summarization, Question Answering, Machine Reading Comprehension, 4W1H, Text Extraction**

I. Introduction

Many social media, banking, education and other NLP applications require mining relevant important information from large text documents. This necessitates the need for an automatic keyword extraction and a summarization model. Automatic keyword extraction is the process of selecting important words from a text document, which is a primary need for text summarization. While summarization is a task that creates a short abstract of a long text document. With the advent of big data analysis, summarization has become powerful technique which provides a glimpse of the whole document.

Text summarization can be performed in three ways: extractive, abstractive and hybrid text summarization. While abstractive summaries were studied using the neural seqeunce transformation method with a large paired document summarization example dataset [1], extractive summarization provides core information consistently and understanable through the steps of feature extraction, feature enhancement, feature summary and generation [2].

Generally, sentences are written based on main information for which human want to represent. In particular, 4W1H(Who, When, Where, What, How) keywords are categories which contain main information. Therefore, those things would be good sources for neural networks to understand the natural language [3]. However, recent researches in Natural Language Processing (NLP) domain still do not utilize that resources despite its surprising progress [4]. Among NLP tasks, summarization task requires to sum up a long text source into short text with core information, and 4W1H keywords would be beneficial to this process [5]. There are some skills in conventional approaches of summarization: extractions for source text, abstractive generation based on a source text, or their hybrid methods [6], [7]. However, they have limitations that don't specify core information or extract unrelated text from a source text [8].

In this paper, we propose a new 4W1H keywords extraction method and an 4W1H keywords-based summarized sentence generation method. Our proposed model has advantages of extracting core information from a sentence. Also, it is possible to compensate the limitations of the previous methods. We apply our proposal methods to BERT and ELECTRA models to generate a summary which are well-known LMs that ware pre-trained with a large-scale text corpus [9], [10]. We also use Transformer decoder, which generates an output sequence of tokens auto-regressively by self-attention while attending to the whole input representation [11], [12]. In experiments, we indicate our proposed 4W1H keywords extraction model can grasp important information from a source text. Also, we record a significant performance on AI Hub Machine Reading Comprehension (MRC) dataset and show that extracted 4W1H keywords play role as a key point in summarization.

II. Related Work

Summarization studies of NLP have been and are underway in a variety of active ways. The summary is the process in which a text is extracted from the original document and the most prominent sentences are extracted [13].

*https://www.aihub.or.kr/aidata/86

Keyword Extraction The method of automatically extracting keywords from a text has been studied for its summarization [14]. Statistical methods for a keyword extraction focused on non-lingusitic features of the document, which included word frequency in the document, term frequency etc., to create a list of keywords [15]–[17]. Linguistic keyword extraction methods utilized linguistic features for keyword extraction which incorporates lexical, syntactic and discourse analysis. [18]–[20]. Machine learning approaches consider key word extraction as a learning problem which requires manually annotated training data. The document whose keywords are to be extracted is given an input to the model which extracts keywords based on training data.

Conventional Summarization Verma et al. (2017) provides core information consistently and understandable through the steps of feature extraction, feature enhancement, feature summary and generation [2]. In this study, summary sentence was constructed by extracting the most relevant sentences using centroid and jaccard similarity. It did not reflect the context and was affected by the frequency in extracting the keywords of a text. In Liu (2019)'s paper, several summarization related hierarchies were constructed by obtaining sentence vectors from BERT [21]. To extract the summaries, capturing document-level features was used by stacking them on top of the output of BERT. On the other hands, the abstractive approaches generate important summarized sentences based on feature states derived from a source text. Recent abstractive research proposed multiple timescale based abstractive summarization model which deals with the presence of multiple compositions, and showed better performance compared to the approaches [22].

III. PROPOSED MODEL

Recently, many pre-trained LMs on large-scale corpus are proposed and showed significant progress in NLP tasks, such as SQuAD, GLUE, and etc [9]. As shown in Fig. 1, BERT, which has an encoder architecture of Transformer, presents intensive solutions that can provide contextualized word representations [9], [12]. It was pre-trained with two unsupervised ways, a reconstructing masked words into original words and a predicting consecutive relation between given two sentences. Based on those ways, BERT learns a distributed representation existing in massive corpus. However, it has limitations which require large number of parameters, computation time, and etc. In order to enhance the limitations, some researchers proposed ELECTRA model which consists of generator and discriminator similar to Generative Adversarial Network (GAN), As shown in Fig 2. ELECTRA is pre-trained in a way that the discriminator to distinguish whether the words produced by the generator are same to original input words. Through this pre-training method, it shows progress performance with smaller resources in NLP domains.

By extension, we design the proposed model using the pre-trained model to grasp the context Transformer, as shown in Fig. 3. It has a formidable force in the filed of deep learning. Transformer is very widely used and has a tremendous impact

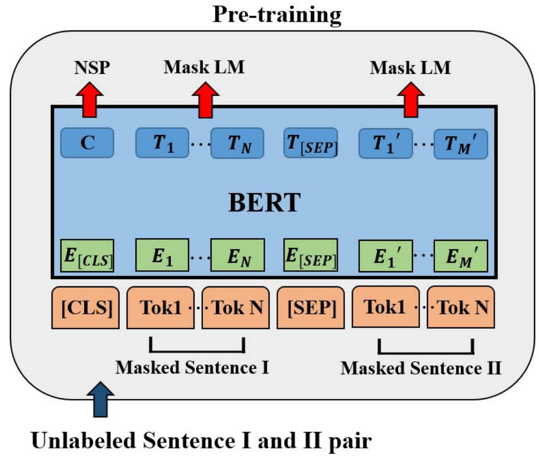

Fig. 1. Pre-training process of BERT model

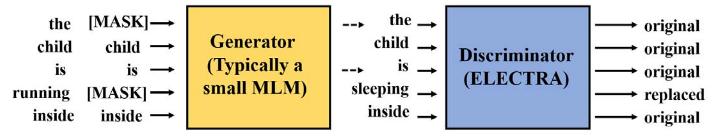

Fig. 2. Pre-training process for replaced token detection of ELECTRA model

on many fields like Natural Language Generation (NLG) [9], [23], [24].

A. 4W1H event extraction

Initially, the types of questions are divided according to 4W1H and use to extract each event. Secondly, in order to extract the answer to a question corresponding to respective event, the question and context are used as inputs of the model. Then the model predicting the span point of the start and end of the answer is used as the base model. Most of the answers extracted here are texts corresponding to words or short sentences. However there may be cases where the answer to a question needs to generate a sentence-based answer that is relatively long. For this reason, it is necessary to additionally construct a model capable of generating abstract sentences as well as extracting text.

Ultimately, we use the BERT, ELECTRA, and Transformer decoder models to construct the proposed model to apply the foresaid [9], [10], [12]. BERT and ELECTRA are pre-trained LMs and fine-tuned according to a task to be performed in this paper. The Transformer decoder is selectively used in the overall proposed model for use in sentence generation tasks.

As a result, this paper presents a new method to generate a summarized sentence using the extracted event information. In this part, it is used to create a new sentence by using a simple rule-based method like the if-then approaches.

IV. EXPERIMENTS

In order to test the 4W1H extraction model based on the pre-trained LMs presented in this paper, we use the MRC dataset

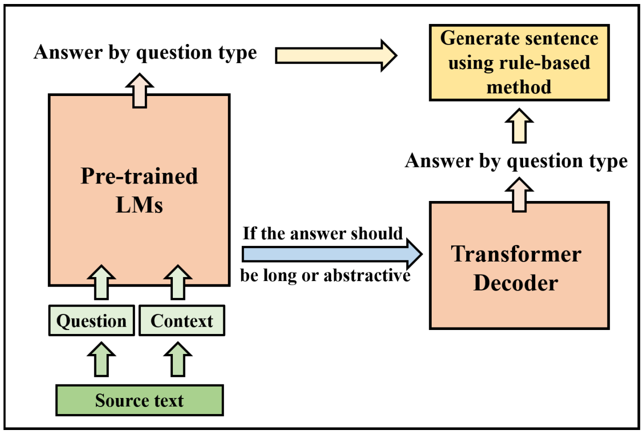

Fig. 3. The overall architecture of the proposed model

TABLE I
PERFORMANCE OF 4W1H EXTRACTION (MEASURE: F1-SCORE)

4W1H	*BERT(%)*	*ELECTRA(%)*
Who	71.21	83.33
When	75.95	86.81
Where	72.32	82.86
What	62.23	87.50
How	63.33	84.14
Average	69.01	84.94

provided by AI Hub. This dataset provides 46,889, 41,622, 34,795, 56,594 and 32,479 for each of 4W1H, respectively. Figure 4 is an example of a machine-readable dataset corresponding to the ”what” type of 4W1H. For each event, training data and validation data are divided in a ratio of 9:1. TITAN RTX is used to learn the proposed 4W1H event extraction model with the batch size as 16. As a result, it takes an average of 5 hours to learn for 3 epochs.

Table I shows the results of performance analysis by event for the 4W1H extraction model. The performance analysis is measured based on F1-score. Unlike the 4W events, the performance of 1H event is measured using the final results of attaching the Transformer decoder to the results of the BERT and ELECTRA. In the results of Table I, when 4W1H events are extracted based on BERT, we get the average performance as 69.01%, and when events are extracted based on ELECTRA, we obtain the average performance as 84.94%. As a result, the ELECTRA model shows 15.92% higher performance than the BERT-based model. The summarized sentence which is presented in Figure 5 is an example of extracting the entire article for each event and then displaying a new summary sentence. This summarized sentence presents a simple summary in a long text and contains the content of main events that must be grasped in the text.

V. CONCLUSION

In this paper, we propose a deep learning model that extracts and provides 4W1H event information in order to detect important events in text when given text. The final result is that the extracted event information is combined through rule-based learning and provided as a summary sentence for a long text.

Context : The era of the Fourth Industrial Revolution. What will future agriculture look like? Even without soil and sunlight, fruits grow, water themselves, and fertilizers! It's becoming a reality, not an imagination. Um Jin-ah will guide you to a state-of-the-art farm that incorporates artificial intelligence technology. [Report] Mushrooms grow on shelves with layers. An enclosed area where LED lights operate instead of sunlight, instead of wind. By controlling temperature, humidity, and amount of carbon dioxide, we increased production rate by 40%. It's run by an IT company, not a farmer. Artificial intelligence-controlled farms! Vinyl house covers open and close themselves. When it is dry, artificial intelligence sprays water on its own, and if it lacks nutrients, it also gives fertilizer. You can even remotely control it with your smartphone.[Ryu Pil-yong/Farmers: When it rains after going out, I had to crack down on the house, but now I can operate it anywhere with a smartphone] These smart farm farms are more than doubling every year. The biggest reason why advanced farming methods are attracting attention is that stable production is possible even in rapidly changing climate conditions. Production increased by 30 percent compared to normal farms and labor costs were reduced by nearly 10 percent. It's cheaper, it's longer available for consumers. The government plans to provide 76 billion won to the smart farm business this year and nearly double the size by 2022 to [Cho Yong-hyun / Head of Customer Service Team at Large Mart: General Strawberries will be operated until early April, but Smart Farm Strawberries will be operated by early June. This is Eom Jin-ah from KBS News.

Question : What did you increase by 40 percent by controlling the temperature, humidity, and amount of carbon dioxide in mushroom cultivation?

Answer : Production rate

Fig. 4. An example of MRC dataset

The proposed model extracts 4W1H information based on the pre-trained LMs and compares the performance results for each of the five events using two representative pre-trained LMs. A study on extracting events of the sixth principle similar to this study has been conducted before, but studies on event extraction using deep learning have not been conducted much. In order to help understand the situational awareness, the 4W1H extraction model proposed in this study is expected to provide timely and accurate information without reading long texts directly. In addition, when generating extracted sentences in the future, we will design to enable End-to-End by using a deep learning-based NLG (e.g. GPT) method rather than the rule-based format currently used [25]–[27].

Context : The Korean government has expressed its intention to respond to the US government's move to impose tariffs on steel. In order to expand the export market, the plan is to decide whether to join the Pacific Rim Economic Partnership Agreement, a free trade agreement involving 11 countries including Japan, in the first half of the year. Reporter Kim Kyung-jin's report. [Report] Deputy Prime Minister Kim Dong-yeon reaffirmed his position to do everything the government can with regard to the US move to impose steel tariffs at the Foreign Economy Ministers' Meeting today. [Kim Dong-yeon / Deputy Prime Minister of Economy : We plan to fully respond to US tariff measures by utilizing all available channels of the government.] Information sent a letter to US Treasury Secretary Manusin to actively persuade the need for tariff exemption on Korean steel. I've sent it, and I'm going to discuss this at next week's G20 Finance Ministers' Meeting. The government also agreed to expand the export market while the United States actively responds to the protection trade policy. Deputy Prime Minister Kim emphasized that he would shape the new northern and new southern policies and actively pioneer the Middle East and Latin American markets … The government has also decided to negotiate a balanced outcome based on the principle of first priority for Koreans regarding the Korean-US FTA, which marks its 6^{th} anniversary.

[4W1H event extraction]

- **who :** Our government

- **when :** Next week

- **where :** G20 Finance Ministers' Meeting

- **what :** US government's move to impose steel tariffs

- **how :** Plan to take full action by using all available government channels

[Summarization sentence using 4W1H event]

At the G20 Finance Ministers' Meeting next week, our government plans to respond to the US government's move to impose tariffs on steel, using all available government channels.

Fig. 5. Sentence generation using 4W1H extraction results

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